An exact certificate-based enumeration of Rolle words of degree six

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Most of the mathematical development, code, certificates, and manuscript were generated using ChatGPT (OpenAI), under the direction of the human author.

Abstract

For a strictly hyperbolic real polynomial whose nonconstant derivatives have mutually distinct roots, record the derivative orders of all roots from left to right. We report a computer-assisted classification of these Rolle words in degree six: exactly 2706 words are realizable, counting reflections separately. The computation starts from 33592 formal interlacing words. The known quintic classification and a theorem of Kostov–Shapiro reduce this to 13404 candidates. Integer realization certificates and 53 exact exclusion stages produce an exhaustive, disjoint classification. Verification uses integer and rational arithmetic; floating-point searches are not used as impossibility proofs. A readable enumerator and an explicit proof-stage ledger accompany the certificate corpus. Most of the mathematical development, software, certificate construction, and initial exposition were generated using ChatGPT in an interactive project directed by the author. This substantial contribution and the absence of an independent human audit are documented explicitly.

Keywords: hyperbolic polynomial; Rolle word; real roots; derivative; exact enumeration; computer-assisted proof; Bernstein basis.

1 Problem, result, and provenance

Let $P \in \mathbb{R}[x]$ have degree n and n distinct real roots. By Rolle's theorem, $P^{(j)}$ has $n-j$ distinct real roots for $0 \le j < n$. We impose the additional general-position assumption that roots belonging to different derivatives are also distinct. Thus the complete collection contains $N = n(n+1)/2$ distinct real numbers.

Definition 1.1. The *Rolle word* $w(P)$ is the word of length N obtained by sorting this collection increasingly and replacing a root of $P^{(j)}$ by j. Let a_n denote the number of realizable words. A word and its reversal are counted separately.

The question belongs to the study of root configurations associated with Arnold and developed by Kostov and Shapiro [1, 2]. Kostov's Theorem 2 [1] gives $a_5 = 116$, whereas 286 formal interlacing words exist in that degree. Kostov–Shapiro [2] establish additional restrictions beyond Rolle interlacing. We use these published results explicitly; the completeness of the quintic list is not inferred from a random search.

Theorem 1.2 (Computer-assisted enumeration)**.** *Under the definition above,* $a_6 = 2706$.

The proof consists of the mathematical reductions described below and the associated exact certificate verification. It is not a computation of connected components of coefficient space. In particular, no assertion that the realization set of a fixed word is connected is required. The present paper makes no separate claim that a literature search has established publication priority for the numerical result or every individual inequality.

Status of verification. The complete archived verifier and the refactored certificate pipeline have been executed. The former is retained as a reference implementation. Agreement of two organizational implementations does not constitute an independent human audit: both were produced within the same AI-assisted project and share the mathematical certificate checkers. The result is not a formal proof in a foundational proof assistant. These qualifications are relevant to the evidential status of Theorem 1.2.

2 Exhaustive enumeration without root finding

Write $x_1^{(j)} < \cdots < x_{n-j}^{(j)}$ for the roots of $P^{(j)}$. The adjacent interlacing relations are

$$x_k^{(j)} < x_k^{(j+1)} < x_{k+1}^{(j)} \quad (0 \le j < n-1,\ 1 \le k < n-j). \tag{1}$$

Let c_j be the number of occurrences of j in a word prefix.

Lemma 2.1. *A word with multiplicities $n, n-1, \ldots, 1$ encodes a linear extension of* (1) *if and only if every prefix satisfies*

$$0 \le c_j - c_{j+1} \le 1 \qquad (0 \le j < n-1). \tag{2}$$

Proof. After deleting every label except j and $j+1$, strict interlacing says that the remaining word alternates, beginning and ending with j. This is equivalent to (2) for that pair. Applying this to every adjacent pair gives exactly all relations (1). The occurrence number of a label fixes its root index, so a word cannot encode two different linear extensions. □

This yields a short depth-first enumerator: append any label whose multiplicity is not exhausted and whose addition preserves (2). There is no numerical feasibility test in this step. The number of completions can also be computed by the recurrence

$$T(c) = \sum_{j \text{ admissible at } c} T(c + e_j), \qquad T(n, n-1, \ldots, 1) = 1. \tag{3}$$

The implementation memoizes this recurrence and separately generates the words. Their equality is tested against the archived partially ordered set enumerator, word by word, for degrees 1 through 6.

Degree n	1	2	3	4	5	6
Formal words	1	1	2	12	286	33592
Realizable words	1	1	2	10	116	2706

The last entry is the result of the full certificate computation, not of the recurrence alone.

2.1 Published necessary restrictions

Removing every zero from a sextic word and subtracting one from the other labels gives the word of P'. The 116 stored quintic words all have exact realization certificates. Kostov's theorem [1, Theorem 2] therefore proves completeness of this particular list. Keeping only sextic words with one of these tails leaves 13950 words.

For a degree-m strictly hyperbolic polynomial f, Theorem 1 of [2] implies

$$\left(x_{j+1}^{(0)} < x_j^{(2)} \text{ for } 1 \le j \le m-2\right) \implies x_2^{(m-3)} < x_1^{(m-1)}. \tag{4}$$

Apply this to every applicable derivative and to its reflection. Exactly 13404 words survive. Both filters are necessary conditions, not realizability tests.

3 Exact realizability certificates

A stored witness consists of an integer polynomial, either given by its coefficients or by distinct integer roots, a positive integer q, and integers $b_0 < \cdots < b_N$. The ith letter j asserts that $P^{(j)}$ has a root in $(b_i/q, b_{i+1}/q)$.

Lemma 3.1. *If the assigned derivative has opposite nonzero signs at the two endpoints of every interval, then the witness realizes the asserted word.*

Proof. The intermediate value theorem gives a root in every assigned interval. For $P^{(j)}$ there are exactly $n-j$ such disjoint intervals, equal to its degree. Therefore it has precisely one root in each, every such root is simple, and there are no additional roots. Interiors of the intervals for all derivative orders are mutually disjoint; no root is an endpoint of its assigned interval. Hence all roots are distinct and have the specified total order. □

If f has degree d, the sign of $f(b/q)$ is the sign of the integer $q^d f(b/q)$. A denominator-clearing Horner evaluation computes this integer directly. Thus the verification requires neither approximate roots nor a tolerance. The new checker additionally rejects duplicate JSON keys, malformed multiplicities, noninteger data, zero denominators, unordered endpoints, and missing sign changes.

The verified sextic corpus has 2706 distinct words and is invariant under reversal. This proves $a_6 \ge 2706$ independently of the exclusion calculation.

4 Soundness of the exclusion certificates

The exclusion corpus uses several mathematical reductions, but its finite checks fall into a small number of explicit forms. Necessity of a reduction is part of the mathematical argument; a valid algebraic identity cannot repair an invalid necessary inequality.

4.1 Linear and quadratic identities

Represent the ordered 21 nodes by 20 positive gaps. Root coordinates relative to the common mean are linear forms in these gaps. Equality of the root means of successive derivatives gives linear equations $Eg = 0$. Interlacing and additional root inequalities give strict forms $Bg > 0$.

Lemma 4.1. *If $y^T E = z^T B$ with $z \ge 0$ and $z \ne 0$, the strict system $Eg = 0$, $Bg > 0$ has no solution.*

Proof. The left side evaluated at a putative solution is zero, whereas the right side is strictly positive. □

The coefficients of these certificates are integers. Quadratic variants multiply equality constraints by gap coordinates and combine them with second-moment identities; the positive side is a nonzero nonnegative combination of products of strict positive forms. Every coefficient of the resulting polynomial identity is checked exactly.

For completeness, two root inequalities used in these systems are recorded here. If $x_1 < \cdots < x_m$ are the roots of f and $y_1 < \cdots < y_{m-1}$ those of f', then, for $m \geq 3$,

$$\frac{(m-k)x_k + x_{k+1}}{m-k+1} < y_k < \frac{x_k + kx_{k+1}}{k+1} \quad (1 \leq k < m). \tag{5}$$

Indeed, split $f'(y_k)/f(y_k) = 0$ into terms to the left and right of y_k, then bound each denominator by its nearest endpoint. At least one bound is strict. In addition, for $1 \leq k \leq m-2$,

$$(k+1)\sum_{i=1}^{k} y_i < k\sum_{i=1}^{k+1} x_i, \tag{6}$$

with the reversed inequality for the corresponding rightmost roots. To see this, realize the y_i as the eigenvalues of the compression of $\operatorname{diag}(x_1, \ldots, x_m)$ to the orthogonal complement of the all-ones vector. The minimum trace on a k-dimensional subspace is $\sum_{i=1}^{k} y_i$. Restricting to vectors supported on the first $k+1$ coordinates and having zero coordinate sum gives trace $\frac{k}{k+1}\sum_{i=1}^{k+1} x_i$. Equality would make this proper subspace invariant under the compression, which is impossible for distinct x_i. The characteristic polynomial identity for this compression follows directly from the resolvent identity $f'(z)/f(z) = \sum_i (z - x_i)^{-1}$.

If the roots are centered at their common mean and S_m denotes their sum of squares, coefficient comparison gives

$$S_{m-1} = \frac{m-2}{m} S_m, \qquad \frac{S_m}{m(m-1)} = \frac{S_2}{2}. \tag{7}$$

These are the quadratic equalities used in the initial moment stage. The same ideas apply to root-subset factors, their derivatives, and consistent deletion of roots. Metric cuts include rational consequences of the span bound

$$\operatorname{span}(f') \geq \sqrt{\frac{m-2}{m}}\, \operatorname{span}(f) \tag{8}$$

of Meir–Sharma [3]. Rational constants are rounded in the direction that preserves necessity, and the rounding is checked by integer squares.

4.2 Polynomial sign identities and Bernstein covers

Other stages normalize a low-order derivative and integrate it, leaving a small number of shape parameters and integration constants. A prescribed word fixes signs of values at marked roots. Eliminating the integration constants produces necessary polynomial inequalities.

For a polynomial f on a rational box, substitute $x_i = \ell_i + (r_i - \ell_i)t_i$, $0 \leq t_i \leq 1$, and expand in the tensor Bernstein basis:

$$f(x(t)) = \sum_{\alpha \leq d} b_\alpha \prod_i \binom{d_i}{\alpha_i} t_i^{\alpha_i} (1-t_i)^{d_i - \alpha_i}. \tag{9}$$

The basis functions are nonnegative and sum to one. Hence $f \geq \min_\alpha b_\alpha$ on the entire closed box. All b_α are rational and are evaluated exactly. Binary subdivision certificates record an axis and its two halves at each internal node; a leaf identifies a polynomial with the required sign. The checker verifies coverage as well as leaf signs. A strict required condition can be

contradicted by a non-strict bound, whereas rejecting an allowed equality boundary requires a strict bound. The checkers distinguish these cases.

Unbounded regions are handled by explicit changes of variables, positive coefficient expansions on a tail, or local algebraic identities near a limiting square. A numerical search failure is never an exclusion certificate. The archived derivations and the checker source for every stage are indexed by the public registry.

4.3 Hermite constraints

For a monic real polynomial f of degree m, let s_k denote its Newton sums, with $s_0 = m$. The Hermite matrix is $H_m = (s_{i+j})_{0\le i,j<m}$. For distinct real roots it is a positive definite Vandermonde Gram matrix. Conversely, if a pair of roots is nonreal, real-coefficient interpolation gives a polynomial taking values $\mathrm{i}, -\mathrm{i}$ on that pair and zero at the other distinct roots. Its Hermite quadratic form is negative. Repeated roots make the matrix singular. This proves the strict hyperbolicity criterion used by the Hermite stages. Newton identities, matrix test-vector identities, and sign bounds for their coefficients are checked in exact arithmetic. Low-order matrix tests are only necessary conditions; passing them does not certify a sextic witness.

5 A common normalization for the final exclusions

Multiplication of P by -1 preserves its word, so we may assume a positive leading coefficient. Normalize the two middle roots of P'' to 0 and 1, and write the outer roots as $-u$ and T, where $u > 0$ and $T > 1$. After a positive scaling of P, write

$$R = P'' = (x+u)x(x-1)(x-T), \qquad T = 1+u+2r, \tag{10}$$

$$H = u^2 + u + 2ur - 2r - 1, \qquad P = F - qx + C, \tag{11}$$

$$F(x) = \frac{x^6}{30} - \frac{(1+r)x^5}{10} - \frac{Hx^4}{12} + \frac{uTx^3}{6}, \qquad Q = F'. \tag{12}$$

Let $a < c < d$ be the roots of R', $v_- < v_+$ those of R'', and $\mu = (1+r)/2$ the root of R'''. Reflection permits the orientation $c < \mu$, equivalent to $r > 0$, since

$$R'(\mu) = -r(r+u)(r+u+1). \tag{13}$$

General position excludes equality. A strictly hyperbolic sextic must have negative first critical minimum and positive fourth critical maximum. These elementary sign requirements are used repeatedly.

The final 28 unresolved words from the historical base are excluded by seven checked rules. The first two share a recorded recovery stage; their net contributions are nevertheless counted separately.

Rule	New words	Cover leaves
Negative left cubic value	2	7
Matched left-minimum/right-maximum regimes	2	241 + 530
Negative right maximum	2	34
Positive return at the mean	4	18
Negative inflection values	6	17
First minimum with $H > 0$	6	81
Central dip and first minimum	6	194

Leaf counts describe finite covers, not the number of independent mathematical hypotheses. Tail inequalities and limiting identities are additional checked parts of the relevant rules. We give the final rule in detail, to exhibit the link between a word and its finite certificate.

6 The final central-dip obstruction

Proposition 6.1. *No strictly hyperbolic sextic in the normalization of Section 5 can satisfy all the conditions*

$$\begin{gathered} R(v_-) > 0, \quad R(\mu) > 0, \quad R(v_+) < 0, \\ P'(a) < 0, \quad P(c) < 0, \quad P'(v_+) > 0, \quad P(0) > 0. \end{gathered} \tag{14}$$

Proof. Interlacing and $R''(0) = -2H$ imply

$$H < 0, \qquad 0 < r < 1, \qquad v_+ > 1, \qquad H + 6r > 0. \tag{15}$$

The last implication uses $R''(1) = -12r - 2H < 0$. Since H is increasing in u and $H(1, r) = 1$, we have $u < 1$. If $s_1 < -u$ is the first root of P', the necessary inequalities are

$$\begin{gathered} C > 0, \quad P(s_1) < 0, \quad 0 < q < \min\{Q(-u), Q(1)\}, \quad q = Q(s_1), \\ q > Q(a), \qquad q > F(c)/c, \qquad q < Q(v_+). \end{gathered} \tag{16}$$

The middle strict lower bound follows from $P(c) < 0$ and $C > 0$.

Direct evaluation gives

$$R'(-3u/4) = -\frac{u(6ru + 16r + 3u^2 + 14u + 8)}{16} < 0, \tag{17}$$

$$R'(-u/2) = \frac{u^2(r + u + 1)}{2} > 0, \tag{18}$$

$$Q(-3u/2) = -\frac{9u^4(5r + 7u + 5)}{160} < 0. \tag{19}$$

Consequently $a = -ut$, $1/2 < t < 3/4$, and $s_1 = -uv$, $1 < v < 3/2$. Furthermore $R'(1/2) = r/2 > 0$, so $c > 1/2$. On $(0, 1)$ the function Q increases strictly, and $F(0) = Q(0) = 0$; hence $F(x)/x$ increases strictly. Thus $q > 2F(1/2)$, whereas $q < Q(-u)$. On $0 \le u \le 2/5$, $0 \le r \le 1$, the minimum Bernstein coefficient of $2F(1/2) - Q(-u)$ is $2143/1000000 > 0$. This exact calculation excludes that rectangle and gives $u > 2/5$.

Define the following polynomials in u, t:

$$D = ut(3t - 2) + 2t - 1, \tag{20}$$

$$N = (2ut + 1)\{u(1 - 2t^2) - 2t + 1\}, \tag{21}$$

$$D_N = 2D + N, \qquad T_N = (1 + u)D + N, \tag{22}$$

$$H_N = D(u^2 + u - 1) + N(u - 1), \qquad H_b = H_N/u. \tag{23}$$

The division defining H_b is exact in the polynomial ring. The logarithmic derivative of R at $a = -ut$ gives

$$D = \frac{ut(1 - t)(ut + 1)}{ut + T} > 0, \qquad r = \frac{N}{2D}, \qquad H_N = DH. \tag{24}$$

It follows that

$$D > 0, \quad N > 0, \quad N < 2D, \quad H_b < 0, \quad H_N + 3N > 0. \tag{25}$$

The full necessary parameter region is contained in the closed box

$$[2/5, 1] \times [1/2, 3/4] \times [1, 3/2] \quad \text{in } (u, t, v). \tag{26}$$

Set

$$E(x) = 60DF(x) = 2Dx^6 - 3D_N x^5 - 5H_N x^4 + 10uT_N x^3, \tag{27}$$

$$Q_A = E'(-ut)/u^4, \qquad Q_X = E'(-uv)/u^4, \tag{28}$$

$$G_q = \{E(-uv) + uvE'(-uv)\}/u^5. \tag{29}$$

All displayed quotients are polynomials. We must have $Q_X > Q_A$, $Q_X > 0$, and $G_q < 0$, the last because $P(s_1) = u^5 G_q/(60D) + C < 0$. Nonnegativity of any polynomial in

$$-D,\ -N,\ N - 2D,\ H_b,\ -(H_N + 3N),\ Q_A - Q_X,\ -Q_X,\ G_q \tag{30}$$

therefore yields a contradiction.

Two further alternatives use certified rational root bounds. A constant $c_L \in [1/2, 1)$ satisfying $2DR'(c_L) \geq 0$ uniformly on a parameter box obeys $c_L \leq c$; $c_L = 1/2$ is available without a separate evaluation. Hence nonnegativity of

$$E(c_L)/c_L - E'(-uv) \tag{31}$$

contradicts $q > F(c)/c \geq F(c_L)/c_L$. Similarly a constant $w \in [1, 2)$ satisfying $-2DR''(w) \geq 0$ obeys $w \leq v_+$; the initial bound $w = 1$ follows from (15). Since Q decreases on $[1, v_+]$, nonnegativity of

$$E'(-uv) - E'(w) \tag{32}$$

contradicts $q < Q(v_+) \leq Q(w)$.

The attached certificate partitions (26) into 194 rational boxes, using 387 tree nodes and maximum depth 28. For the ten alternatives in (30), (31), and (32), respectively, the leaf counts are

$$(0, 3, 3, 5, 3, 63, 0, 72, 22, 23). \tag{33}$$

The checker reconstructs the polynomials from E, verifies the rationalization identities, checks each auxiliary root bound, and checks every Bernstein coefficient with rational arithmetic. All of (26), including its boundary, is covered. Every possible parameter point therefore violates a necessary strict inequality. □

The six final words satisfying (14) are

```
01021320413052410321 0 01230142503140231201 0
01021320413052413021 0 01203142503140231201 0
01021320413052413201 0 01023142503140231201 0
```

Each row is a reflection pair. The predicate is independently checked by reading derivative signs directly from word positions. It matches 390 initial candidates, of which 384 had already been excluded; its net contribution is exactly six.

7 Set accounting and the proof of the count

Let C_0 be the 13404 words left after the published filters. For stage i, let M_i be the set recognized by its checked obstruction. The actual exclusion and residual sets are defined by

$$E_i = C_{i-1} \cap M_i, \qquad C_i = C_{i-1} \setminus E_i. \tag{34}$$

The sets E_i are disjoint by construction. A rule can match many words already excluded without changing its contribution. The program checks that no M_i intersects the verified witness set, and that the final residual set equals that witness set exactly, not merely in cardinality. For historical predicate stages the individual exclusion lists are also compared with their archived counterparts. Expected counts serve only as regression checks and are never used to select excluded words.

Proof of Theorem 1.2. Lemma 2.1 enumerates every formal word exactly once. The quintic theorem and (4) leave C_0, which contains every realizable sextic word. Each checked exclusion rule is a necessary-sign contradiction of the kinds described above, so no realizable word is removed by (34). The 53 stages remove 10698 distinct words and leave precisely the 2706 words with integer realization certificates. Lemma 3.1 proves all these realizations. Thus both bounds are 2706. □

The full incremental ledger is given in Appendix A. The historical base ended at 2734, followed by seven contributions $2, 2, 2, 4, 6, 6, 6$. The intermediate bound is retained in old files for reproducibility and must not be mistaken for the current result.

8 Software structure and reproduction

The readable front end is the package `rolle6`. Its responsibilities are separated as follows:

Module	Responsibility
`words.py`	Prefix enumeration, dynamic-programming counts, tail extraction, and published filters.
`witnesses.py`	Integer evaluation and exact interval witness validation.
`stages.py`	Explicit registry of 53 proof stages and their reference increments.
`backend.py`	Invocation of the archived exact proof checkers, with a per-run cache.
`pipeline.py`	Proof-before-exclusion control flow, set ledger, input hashes, and machine-readable results.

The specialized mathematical modules in `research/` remain unchanged. Their hashes are compared with the completed pre-refactoring archive. This preserves the proof corpus while replacing the enumeration and monolithic bookkeeping. It is not a claim that every algebraic checker has been independently reimplemented.

From the root of the ancillary code directory, use Python 3.10 or later:

```
python3 -m rolle6 enumerate --degree 6
python3 -m rolle6 verify --output-dir verification_output
python3 -m unittest discover -s tests -v
```

No third-party Python package is required for verification. A complete run writes the proof log, each stage's exact exclusion list, the final realizable list, and a dependency hash inventory. The verifier refuses optimized Python execution, because the archived checkers use assertions that `python -O` would disable.

The refactoring tests compare every generated word with the earlier partially ordered set enumerator in degrees 1 through 6, compare every published-filter decision, check all 2822 quintic and sextic witnesses, and check rejection of deliberately damaged certificates. The entire exclusion pipeline is then run against the archived data. Search scripts, random seeds, floating-point optimizers, and intermediate unsuccessful experiments are not required to reproduce the result.

9 Substantial use of ChatGPT and verification limits

Most of the mathematical development and computational work in this project was generated by ChatGPT (OpenAI), not merely edited or translated with it. This includes proposing many of the reductions, drafting most proof arguments, writing and revising the search and verification programs, constructing exact certificates, analyzing the remaining cases, performing the code refactoring, and drafting most of this manuscript. The

declaration is intended as substantive attribution of the tool's role, rather than a generic language-editing acknowledgement.

The human author selected the research problem, directed the interactive investigation, proposed and questioned lines of attack, requested checks against overlap and missed cases, imposed the incremental stopping and saving procedure, and requested this transparent account of the work. The project was carried out through ChatGPT during November 2026. Exact model-version attribution is not available for every part of the extended session; no unsupported per-step model identifier or contribution percentage is asserted.

ChatGPT is disclosed as a research and writing tool. The human author remains responsible for any submitted version. In particular, repeated checks produced within the same AI-assisted workflow are not a substitute for an independent mathematical and software audit. At the time of this preprint, no such independent human audit of the entire chain has been completed. Publishing the exact certificates, explicit assumptions, readable enumeration, per-word ledger, and unchanged verification backend is intended to make that audit feasible.

Three distinct issues remain separate: execution of the exact checker, correctness of the mathematical reductions implemented by it, and publication novelty. The computations establish the first and accompany the arguments for the second; they do not by themselves settle the third. The manuscript does not attribute classical results, the known quintic classification, or published root inequalities to the AI-assisted project.

A Complete incremental exclusion ledger

The initial candidate count is 13404. Each row lists only newly removed words. The machine-readable registry supplies the corresponding checker and explanatory source; the verification report supplies the words themselves.

No.	Rule	New	Remaining
1	Linear root inequalities	2718	10686
2	Quadratic root moments	2282	8404
3	Products of positive linear forms	4	8400
4	Quintic primitive types	1168	7232
5	Primitive sign at the mean	396	6836
6	First and last interval averages	534	6302
7	Middle interval and convexity	558	5744
8	Central-value identities	298	5446
9	Parametric middle-root identities	212	5234
10	Polar derivative at the mean	12	5222
11	Deletion of one root	44	5178
12	Joint deletion of roots	8	5170
13	Subfactor sign constraints	16	5154
14	Metric span constraints	6	5148
15	Coupled second moments	2	5146
16	Quartic middle-root bridge	82	5064
17	Outer cubic roots	6	5058
18	Two-parameter polynomial bounds	6	5052
19	Right critical maximum	12	5040
20	Stationary-point patterns	1152	3888
21	Inflection-point patterns	790	3098
22	Full stationary constraints	30	3068
23	Covering boundary charts	16	3052
24	Left minimum and inflection	2	3050
25	Coupled root constraints	142	2908
26	Pairs of inflection points	38	2870

No.	Rule (continued)	New	Remaining
27	Hermite fibres	8	2862
28	Hermite endpoint bounds	10	2852
29	Laguerre integral identities	6	2846
30	Outer secants	44	2802
31	Marker quadrature	2	2800
32	Hermite marker matrices	2	2798
33	Order-six Hermite markers	4	2794
34	Marker boundary charts	14	2780
35	Moving Hermite markers	4	2776
36	Quintic minimum envelope	16	2760
37	Three inflection constraints	2	2758
38	Middle triple identity	4	2754
39	Right fourth-derivative minimum	2	2752
40	Outer inflection average	2	2750
41	Central cubic chord	4	2746
42	Left cubic chord	2	2744
43	Left cubic and mean average	4	2740
44	Fourth-derivative average	2	2738
45	Fourth-derivative chord	2	2736
46	Outer cubic chord	2	2734
47	Negative left cubic value	2	2732
48	Matched extrema regimes	2	2730
49	Negative right maximum	2	2728
50	Positive return at the mean	4	2724
51	Three negative inflection values	6	2718
52	First minimum with positive H	6	2712
53	Central dip and first minimum	6	2706

B Minimal enumeration pseudocode

The following equivalent form emphasizes the prefix invariant. The actual implementation also checks input types and supports a memoized counting mode.

```
def enumerate_words(n):
    target = tuple(range(n, 0, -1))

    def visit(counts, prefix):
        if counts == target:
            yield prefix
            return
        for j in range(n):
            if counts[j] == target[j]:
                continue
            updated = list(counts)
            updated[j] += 1
            if all(0 <= updated[k] - updated[k+1] <= 1
                   for k in range(n-1)):
                yield from visit(tuple(updated), prefix + str(j))

    yield from visit((0,) * n, "")
```

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